

An object is either at rest or moves with a constant velocity (in **equilibrium**) unless acted upon by a **resultant** force.

1. In each of the following cases decide whether the forces acting on the object are in equilibrium.
 - a. A car that has been stationary, as it moves away from a set of traffic lights.
 - b. A parachutist descending at a constant rate.
 - c. A rock falling freely.
 - d. A book resting on a table.
 - e. A plane flying at a constant speed in a straight line but losing height at a constant rate.
 - f. A car going around a corner at constant speed.

2. In each of the following situations draw a force diagram to show all of the possible forces acting **on the driver** who is sitting in an open-topped sports car on a non-windy day. **Label each force.**

In each case state whether the forces acting forward are: **greater than, equal to or less than** the forces acting backwards.

- a. The car is stationary.
 - b. The car is moving with constant velocity.
 - c. The car is accelerating.
 - d. The car is slowly decelerating.
 - e. The car decelerates suddenly.
3. Comment on the following statements.
 - a) An object in motion is having a force applied to it.
 - b) A ball falling out of an airplane would fall straight down to the ground
 - c) Which object would be experiencing a greater force: A ball travelling at 20 m/s or the same ball travelling at 10 m/s
 - d) Being hit by a ball travelling at 20 m/s will produce more force on you than the same ball travelling at 10 m/s
4. A box sits on my desk and I push it with my finger, exerting a horizontal force of 30 N on it. The box does not move.
 - a. What is the frictional force of the table on the box?
 - b. I now push with a force of only 20 N on the box.
 - (i) How will the box move now?
 - (ii) What is the frictional force of the table on the box now?
 - c. Sketch a graph to show how the Frictional Force of the table on the box (vertical axis) varies as a function of the Force Applied to the box (horizontal axis).